Set notation

1 List the elements of the following sets:

a $S = \{\text{numbers that can be rolled on a standard die}\}$

b $S = \{\text{positive odd numbers less than } 10\}$

2 Consider the sets: $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{1, 2, 3, 4\}$. If there are no elements contained outside of these sets, find B'.

3 Consider the following sets:

- $A = \{\text{people who like football}\}$
- $B = \{\text{people who like softball}\}$
- $C = \{\text{people who like swimming}\}$
- $D = \{\text{people who do not like any of these}\}$

a Describe set B'.

b Describe set D'.

4 State whether the following are true:

a $\{9, 6, 7, 3, 4\} = \{7, 4, 3, 6, 9\}$

b $\{7, 4, 5, 8\} = \{5, 4, 0, 8, 7\}$

Intersection and union

5 Describe the meaning of the following:

a Complement of set A.

b The union of set A and set B.

c The intersection of set A and set B.

6 For each group of sets, list the elements of "either A or B":

a $A = \{1, 2, 4, 8, 16\}, B = \{1, 2, 4, 8\}, C = \{1, 2, 4, 8\}$

b $A = \{2, 4, 6, 8, 10, 12, 16\}, B = \{1, 3, 5, 7, 11, 13, 15, 17, 21\}$

7 Consider the sets:

- $A = \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100\}$
- $B = \{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$

List the elements found in both A and B.

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8 For each group of sets, list the elements of $A \cap B$:

a $A = \{\text{even numbers}\}$ and $B = \{\text{square numbers less than } 100\}$

b $A = \{\text{multiples of } 5\}$ and $B = \{\text{positive numbers less than } 50\}$

c $A = \{3, 4, 5, 6, 7\}$ and $B = \{3, 7, 8, 9\}$

9 P and Q are sets of car colours:

- $P = \{\text{red, white, yellow}\}$
- $Q = \{\text{white, blue}\}$

There are no other car colours in universal set U. State whether the following are true:

a The set P' is the set $\{\text{blue, white}\}$

b $P \cup Q$ is a set consisting of blue, red, white, yellow and two more elements.

10 P and Q are sets of vegetable types:

- $P = \{\text{carrots, cauliflowers, beans}\}$
- $Q = \{\text{cauliflowers, potatoes}\}$

There are no other vegetable types in universal set U. State whether the following are true:

a $P \cup Q$ is the set of all vegetable types.

b $P \cap Q$ only contains the element 'cauliflowers'.

11 For the following sets, list the elements of:

i $B \cup A$ **ii** $A \cap B$

a A is the set of factors of 24, and B is the set of factors of 36.

b A is the set of prime numbers less than 40 and B is the set of positive integers less than 12.

12 If set A is the set of possible outcomes from rolling a standard die, and set B is the set of possible outcomes from rolling an eight-sided die, list the elements of:

a Set A **b** Set B **c** $A \cap B$ **d** $A \cup B$

13 For the following sets, describe:

i $A \cup A'$ **ii** $A' \cap B'$

a Let A be the set factors of 24, and B be the set factors of 36.

b Let A be the set multiples of 4 less than 70, and B be the set multiples of 6 less than 70.

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14 Let A be the set of multiples of 4 less than 70, and B be the set of multiples of 6 less than 70.

a Describe $(A \cap B)'$.

b Describe $A' \cap B'$.

c Is $A' \cap B'$ the same as $(A \cap B)'$?

15 Let A be the set of multiples of 4 less than 70, and B be the set of multiples of 6 less than 70.

a Describe $(A \cup B)'$.

b Describe $A' \cup B'$.

c Is $(A \cup B)'$ the same as $A' \cup B'$?

16 Let $P = \{\text{roses, lillies, daisies}\}$ and $Q = \{\text{lillies, sunflowers}\}$, and suppose that there are no other flower kinds in universal set U. Is $(P \cup Q)'$ empty?

Venn diagrams

17 Draw a Venn diagram to represent the following sets:

a $X = \{\text{magenta, yellow, cyan}\}$
$Y = \{\text{ruby, cyan, yellow}\}$

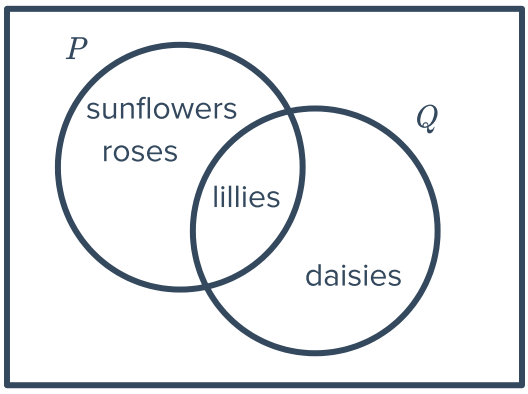
b $A = \{9, 12, 15, 18\}$
$B = \{10, 15, 20\}$

c $B = \{\text{England, Scotland, Ireland, Northern Ireland, Wales}\}$
$U = \{\text{Northern Ireland, England, Scotland, Wales}\}$

18 P and Q are sets of flower varieties:

- $P = \{\text{roses, lillies and daisies}\}$
- $Q = \{\text{lillies and sunflowers}\}$

There are no other flower varieties in universal set E. Is the given venn diagram representing these sets?



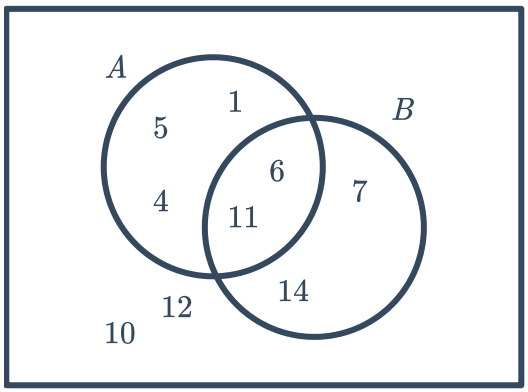
19 How many regions are there in a Venn diagram with two overlapping sets?

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20 We are interested in the colour of a card randomly drawn from a standard deck. Draw a Venn diagram to illustrates this.

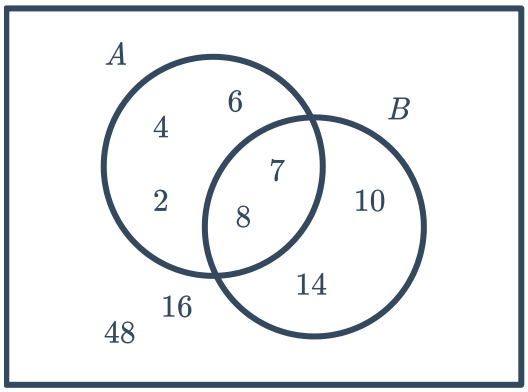
21 Consider the Venn diagram:
List the elements in:

a A' **b** B'



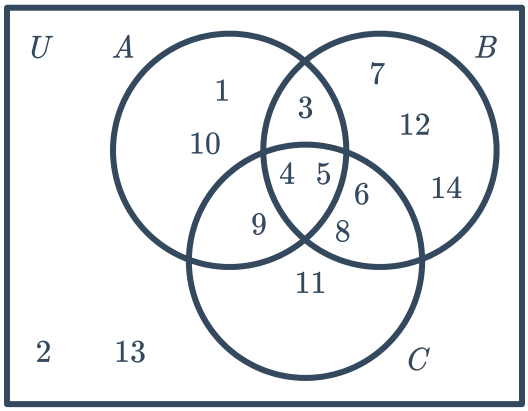
22 Consider the Venn diagram:
List the elements in:

a $A \cap B$ **b** $A \cup B$



23 Consider the Venn diagram:
List the elements in:

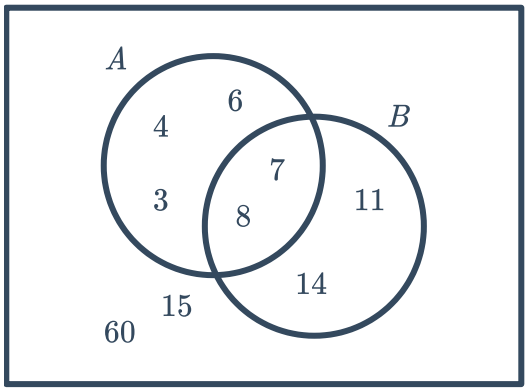
a A **b** U **c** B'



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24 Consider the Venn diagram:
List the elements in:

a $A \cap B'$ **b** $(A \cup B)'$

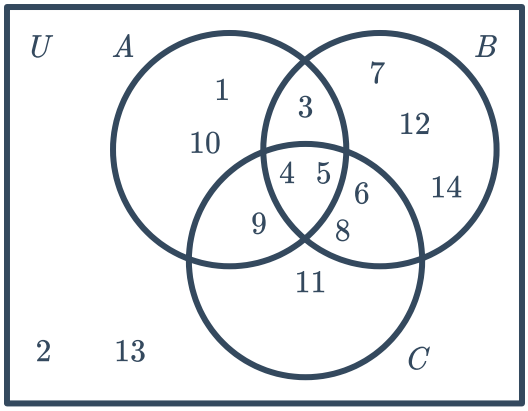


25 Consider the Venn diagram:
List the elements in:

a $A \cap C$

b $(B \cap C)'$

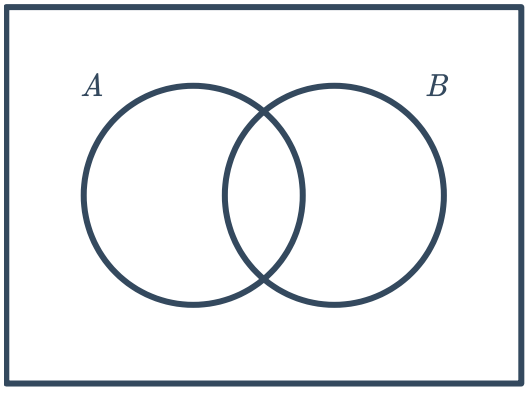
c $A \cap B \cap C$



26 Consider the given Venn diagram and state whether the following statements are equal for all sets A and B:

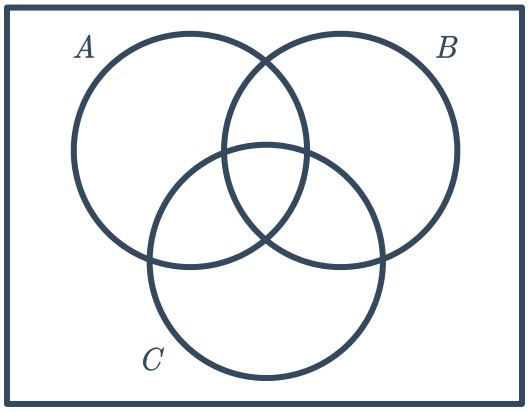
a $(A \cap B)'$ and $A' \cup B'$

b $A' \cap B'$ and $A \cup B'$



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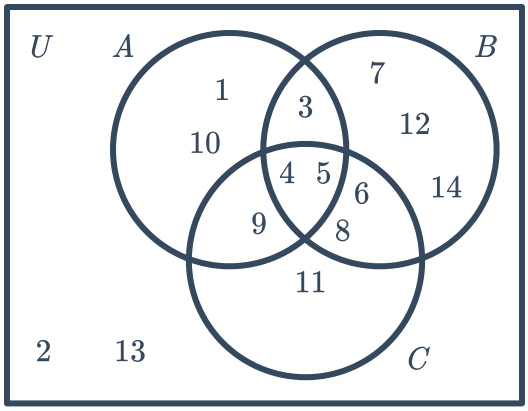
27 Consider the given Venn diagram and state whether the following statements are equal for all sets A, B and C:
$A \cap (B \cup C)'$ and $(A \cap B) \cup C$



28 Consider the Venn diagram and list all of the items in:

a $A \cap (B \cup C)$

b $(A \cap B)'$



29 Consider the Venn diagram and find the value of the following:

a $A \cap B' \cap C'$

b $A \cap B \cap C'$

c $A' \cap B \cap C'$

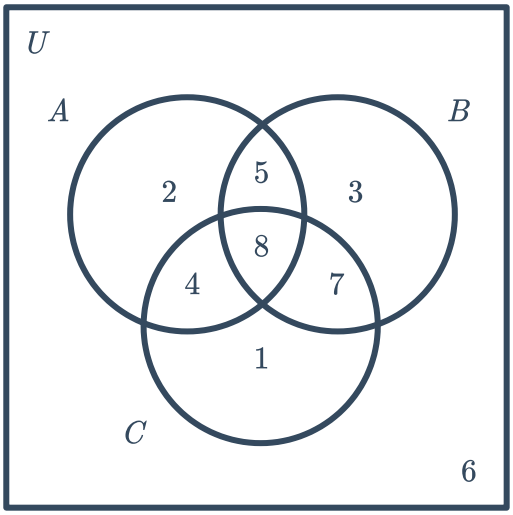
d $A \cap B' \cap C$

e $A \cap B \cap C$

f $A' \cap B \cap C$

g $A' \cap B' \cap C$

h $A' \cap B' \cap C'$



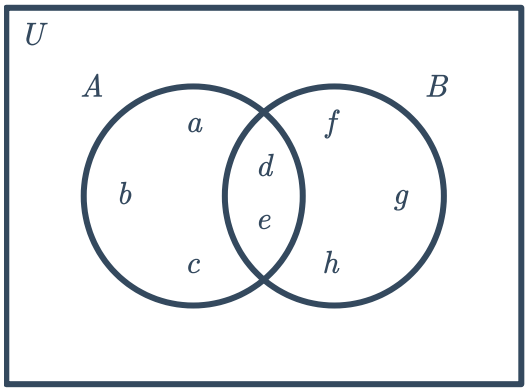
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30 Consider the formula $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

a Does the equation hold for the sets $A = \{a, b, c, d\}$ and $B = \{b, d, e, f, g, h\}$?

b Does the equation hold for the sets $A = \{2, 4, 5, 6, 8, 9\}$ and $B = \{0, 1, 3, 4, 5, 7, 10\}$?

c Explain why the formula holds for the sets in the Venn diagram.



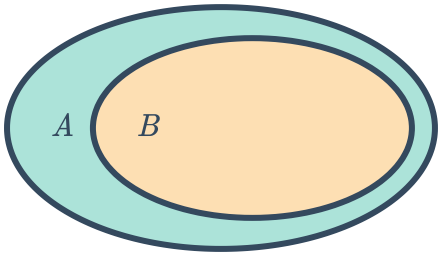
31 The result of a recent survey showed that 34 people own a dog, 33 own a cat, and 13 own both a dog and a cat. How many people surveyed own at least one dog or cat?

32 The results from a recent survey showed that 79 people speak Spanish or French. Of these, 49 speak Spanish, and 18 speak both Spanish and French. Find the number of people surveyed who speak French.

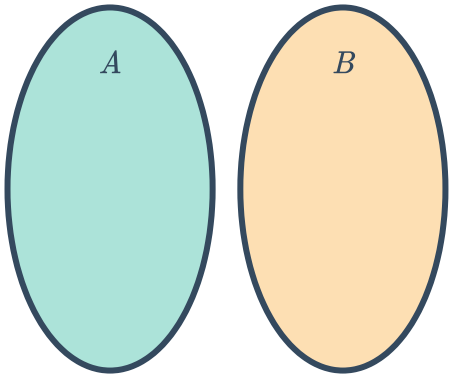
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33 For each of the following, identify the Venn diagram that best represents the sets A and B:

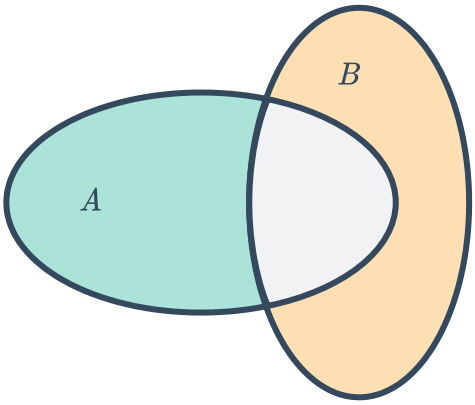
Venn diagram 1



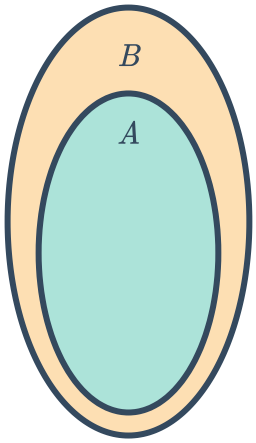
Venn diagram 2



Venn diagram 3



Venn diagram 4



a $A = \{\text{Earth}\}$, $B = \{\text{Planets}\}$

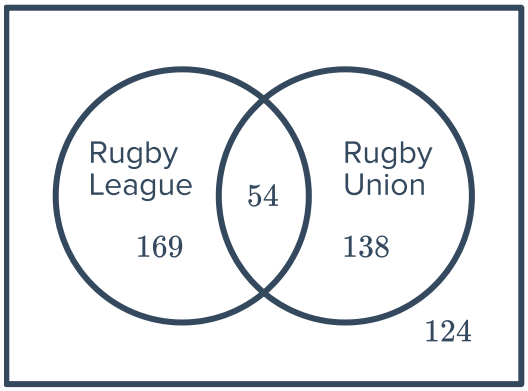
b $A = \{6, 8, 4, 2, 7\}$, $B = \{3, 6, 1\}$

c $A = \{\text{w}, \text{i}, \text{n}, \text{d}\}$, $B = \{\text{e}, \text{a}, \text{r}, \text{t}, \text{h}\}$

d $A = \{\text{Animals found in Australia}\}$, $B = \{\text{Animals found in NSW}\}$

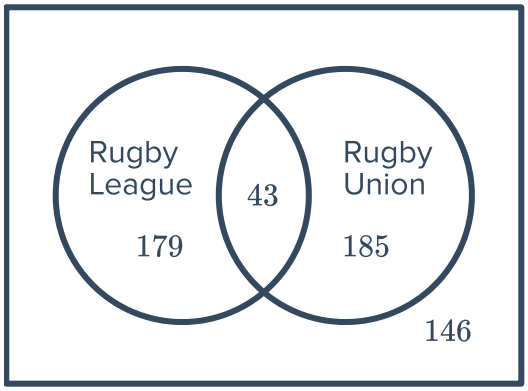
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34 The given Venn diagram shows the number of students in a school playing Rugby League, Rugby Union, both or neither:
Find the number of students who:



- **a** Play Rugby League only.
- **b** Play Rugby League.
- **c** Play Rugby Union.
- **d** Play Rugby Union only.
- **e** Do not play Rugby League.
- **f** Do not play Rugby Union.

35 Consider the Venn diagram:
Complete the table of values below.



	Play Rugby League	Don't play Rugby League
Play Rugby Union		
Don't play Rugby Union		

Mutually exclusive events

36 When picking a random card from a standard pack, which two of the four events listed share no common outcomes?

- **Event A:** picking a black card
- **Event B**: picking a King
- **Event C:** picking a spade
- **Event D**: picking a club

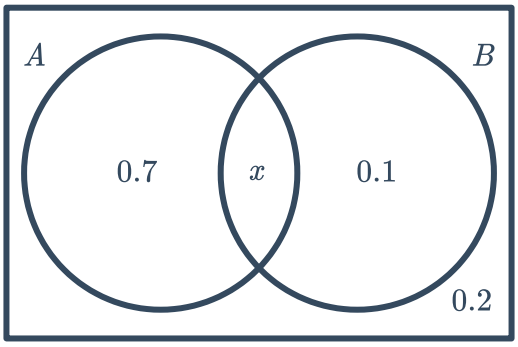
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37 For the following Venn diagrams:

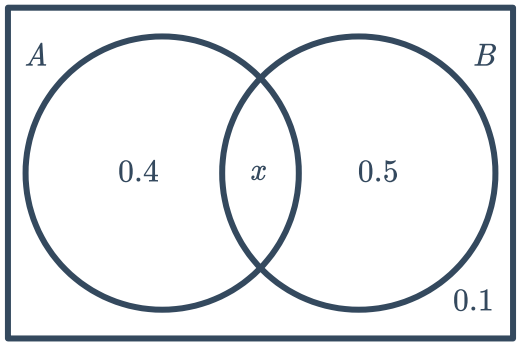
i Calculate the value of x.

ii State whether the events A and B are mutually exclusive.

a



b



38 For each of the following groups of probabilities:

i Find $P(A \cap B')$.

ii Find $P(B \cap A')$.

iii Hence, find $P(A \cap B)$.

iv State whether the events A and B are mutually exclusive.

a $P(A \cup B) = 0.6$, $P(A') = 0.6$ and $P(B') = 0.7$

b $P(A \cup B) = 0.3$, $P(A') = 0.8$ and $P(B') = 0.9$

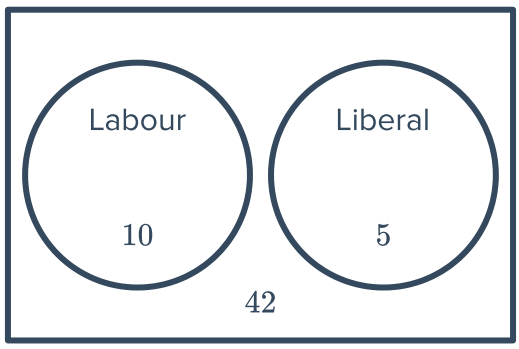
c $P(A' \cap B') = 0.1$, $P(A) = 0.8$ and $P(B) = 0.3$

d $P(A' \cap B') = 0.2$, $P(A) = 0.1$ and $P(B) = 0.7$

39 The Venn diagram shows the party preferences of voters in a sample of the population.

a Find the probability that a random voter has a preference for both Labour and Liberal.

b Are the preferences of voters in this population mutually exclusive?



40 Two events A and B are mutually exclusive. If $P(A) = 0.37$ and $P(A \text{ or } B) = 0.73$, find $P(B)$.

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41 If A and B are mutually exclusive and $P(A) = \frac{1}{4}$ and $P(B) = \frac{1}{2}$, find $P(A \cup B)$.

Non-mutually exclusive events

42 In an experiment, a die is rolled and the number appearing on the uppermost face is noted. Describe $P(A \cup B)$ if:

- Event A = getting an odd number
- Event B = getting a multiple of 3

43 If $P(A) = \frac{1}{5}$, $P(B) = \frac{1}{6}$ and $P(A \cap B) = \frac{1}{30}$, find $P(A \cup B)$.

44 If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{5}$ and $P(A \cup B) = \frac{2}{15}$, find $P(A \cap B)$.

45 If $P(A \cup B) = \frac{7}{15}$, $P(A \cap B) = \frac{1}{15}$ and $P(A) = \frac{1}{3}$, find $P(B)$.

46 If $P(A) = 0.8$, $P(B) = 0.75$ and $P(A \text{ and } B) = 0.6$, find $P(A \text{ or } B)$.

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